

KidneyGrid

Helping strangers swap kidneys so families can save each other

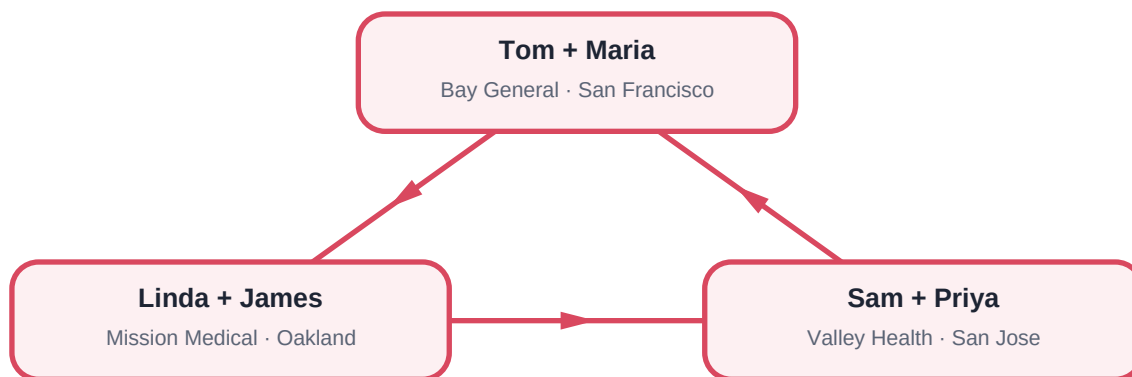
Maresh's project for the Flower Collaborative Agent Hackathon · Stanford · Sept 29, 2026

The problem

More than **95,000** people in the US are waiting for a kidney. In 2024, over **8,000** of them died or became too sick while waiting. Many already have a husband, wife or partner who wants to give them a kidney, but for at least a third of them, the kidney is the wrong match.

The clever fix: swap!

If Tom can't give to his wife Maria, maybe he can give to James. James's wife Linda gives to Priya, and Priya's partner Sam gives to Maria. Everyone gives, everyone receives. The catch: those families are often at **different hospitals**, and hospitals can't share patient records. So these loops are rarely found.



Each donor gives a kidney to a stranger so their own loved one receives one.

What KidneyGrid does

Every hospital gets its own AI helper (an "agent") that keeps its patients' information **locked inside that hospital**. The agents only share anonymous yes/no answers like "this donor could work for one of my patients". A coordinator agent puts the puzzle together, finds the best loops, and doctors approve every match before anyone's name is revealed. It's built on **Flower**, a platform for AI that learns and works together without pooling private data.

It already works

0

matches when each hospital
works alone

3

with simple first-come
matching

5

with KidneyGrid, including
Maria, the hardest match

It ran tonight on a real Flower network of five hospital computers: five transplants found in about 8 seconds, with no patient record ever leaving its hospital. Tomorrow: build the live demo, present to judges from Nvidia, Meta, AMD, ARM and Nebius, and hope to win!